MongoDB Problem Statements

Problem 1: Bookstore System

Problem Statement:

A bookstore system stores book inventory details such as book name, author, category, price, and stock in a MongoDB collection. The store manager wants to view and filter books efficiently.

Questions:

1. Create a books collection and insert one book document.
2. Insert multiple book records using insertMany().
3. Retrieve all books from the collection.
4. Display only bookName and price using projection.
5. Find books where category is "Science", sort them by price in ascending order, and limit the result to 5.

Problem 2: Food Ordering System

Problem Statement:

A food ordering system stores order details such as customer name, item, quantity, total amount, and status. The system frequently updates orders.

Questions:

1. Create an orders collection and insert one order document.
2. Insert multiple order records using insertMany().
3. Update the status of customer "Kiran" to "Delivered".
4. Update all orders where status is "Pending" to "Confirmed".
5. Delete orders where quantity is less than 1 and retrieve orders where totalAmount is greater than 800.

Problem 3: Library Membership System

Problem Statement:

A library membership system stores member details such as member name, membership type, number of books borrowed, city, and status. The admin needs to filter member data based on conditions.

Questions:

1. Create a members collection and insert one member document.
2. Insert multiple member records using insertMany().
3. Retrieve members where booksBorrowed is greater than 3.
4. Find members in "Chennai" with booksBorrowed greater than 2.
5. Display members where status is not "Active", filter by cities (Chennai or Trichy), and sort by booksBorrowed in descending order.

Problem 4: Digital Payment System

Problem Statement:

A digital payment system stores transaction details such as transaction ID, user, amount, payment method, and status. Due to large transaction data, optimization using indexing is required.

Questions:

1. Create a transactions collection and insert sample documents.
2. Create an index on transactionId.
3. Create a compound index on method and amount.
4. Use explain() to analyze a query filtering transactions by method.
5. Identify the best index type and explain how indexing improves performance.

Problem 5: Marketing Analytics System

Problem Statement:

A marketing analytics system stores campaign data such as campaign name, channel, number of leads, and revenue. Management wants insights using aggregation queries.

Questions:

1. Create a campaigns collection and insert sample documents.
2. Calculate total revenue.
3. Find the average number of leads per channel.
4. Group data by channel and count campaigns.
5. Display top 3 campaigns with highest revenue and filter records where revenue is greater than 40000 before grouping.
